1 Configure and compile the C++ population simulator

Precomputed SEVN / MOBSE stellar catalogues + cluster tables + cosmic-age grid
Set input_params.h: paths, candidate counts, cluster physics, spins; build BPOP.x

2 Allocate candidates and sample their formation conditions

Isolated / young / globular / nuclear channels; draw z_{for} and Z, then group by metallicity

3a Isolated binary evolution

Draw a complete upstream binary record:
component masses, orbit and delay; assign spins

4a Use the imported merger delay

$t_{merge} = t_{for} + \Delta t_{input}$
Keep correlations from stellar binary evolution.

Merger before the present age?

Yes

5a Isolated merger properties

Masses, spins, remnant and effective spin

No

Non-mergers

3b Young / globular / nuclear clusters

Draw cluster mass and radius; estimate escape speed.
Sample BHs retained after natal kicks; assign spins.

4b Assemble and evolve a dynamical binary

Birth / sinking → binary formation → pairing / hardening
→ GW inspiral; check ejection versus in-cluster merger.
Update the cluster as time advances.

Merger before the present age?

Yes

5b Calculate the merger remnant

Numerical-relativity fits → mass, spin and GW recoil.
Record the dynamical chain step.

Retained after recoil, and another merger possible?

Yes: reuse remnant + draw companion

No: end chain

6 Archive the C++ catalogues and run configuration

Catalogue.txt
Isolated mergers + selected terminal chain outcomes

Catalogue_multiple_dyn.txt
Intermediate dynamical chain records

Larger_than_tH.txt
Failed / unfinished candidates; efficiency denominators

retention_Z<Z>.dat
Natal-kick cumulative retention tables

Catalogue inputs

7 Set Python cosmological models

Channel formation histories, metallicity probabilities,
normalizations, external isolated efficiency; Planck18.

8 Estimate merger efficiency by metallicity

$\eta_c(Z)$: mergers per $M_\odot$ of formed stars
Isolated: external efficiency table.
Clusters: merger / non-merger yields, BH and binary fractions, and natal-kick retention.

9 Weight formation histories and delays

Sum metallicity bins and integrate formation epochs:
$\mathcal{R}_c(z) = 10^9 \sum_k \int dz_f \psi_c \eta_{c,k} P_{c,k} q_{c,k} |dt/dz_f|$
Implemented delay-bin weight: $q = dN/(N \Delta t_{merger})$.
Output: channel rates $\mathcal{R}_c(z)$ [Gpc^{-3} yr^{-1}].
See the normalization note below.

10 Convert rates to observer counts

$\Delta N_c \simeq \mathcal{R}_c(z) \Delta V_{com} T_{obs}/(1+z)$
Comoving volume × observer interval;
the redshift factor converts source to observer time.

11 Draw mock cosmic event catalogues

Match channel, metallicity and merger-redshift bins;
sample catalogue rows with replacement.
Write channel files and Catalogue_global.txt.

Sparse candidate bins can reduce written counts.
Detector selection requires an additional model.

Reading the diagram: arrows show logical dependencies; candidate fractions set sampling coverage, not cosmic channel fractions. BH = black hole; GW = gravitational wave.

Rate normalization: the implemented delay-bin denominator includes non-mergers, as does the dynamical efficiency; their combined normalization needs scientific review.

Pop III trial: $Z = 10^{-11}$, generic formation histories and zero isolated efficiency; calibrated Pop III rates remain unfinished. Source: BPOP_WORKFLOW.md, §§1–11.